

Evaluation-Target Mismatch in Longitudinal Motor-Imagery BCI: Leave-One-Session-Out and Expanding-Window Validation

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ABSTRACT Cross-session electroencephalography studies often use leave-one-session-out (LOSO) evaluation, which trains on every recorded session except the target. For an early target, this training set contains sessions recorded later in the participant's trajectory. LOSO therefore measures retrospective compatibility with a completed trajectory, whereas a prospective update can use only the labeled history available at the target time. We quantified the resulting evaluation-target mismatch by pairing LOSO with expanding-window validation at identical target sessions in six longitudinal motor-imagery datasets. The audit covered 138 participants and two established decoder families: common spatial patterns with linear discriminant analysis and tangent-space features with logistic regression. Across all prospective origins, LOSO reported an average area under the receiver operating characteristic curve (AUC) that was 0.030 higher. The mean difference was positive under each aggregation rule and for both decoders, with substantial variation across recording designs. The higher-scoring decoder differed between protocols for some participants, primarily where the two decoders had similar scores. A donor-session-count-matched reference partitioned the protocol difference into two descriptive contrasts. Regrouping within-day blocks into recording days preserved a positive conditional difference. Longitudinal BCI evaluation should define the deployment decision and its available information before choosing the split.

INDEX TERMS Brain-computer interfaces, cross-session decoding, electroencephalography, longitudinal evaluation, motor imagery, prospective validation.

I. INTRODUCTION

LONGITUDINAL brain-computer interfaces (BCIs) are evaluated at a sequence of calibration decisions. Before a new session, a decoder can use previously labeled recordings; after the session, newly acquired labels may support another update. An evaluation protocol represents the intended use only when its training information matches what is available at that decision point.

Leave-one-session-out (LOSO) validation withholds one target session and trains on every other recorded session. It is a natural measure of how well a session fits the participant's completed recording trajectory and is available in reproducible BCI benchmarking frameworks [1], [2]. At an early target, however, LOSO also uses later sessions. An expanding-window design trains on sessions observed before the target. The two protocols test the same trials but estimate different operational quantities.

This distinction matters because cross-session research increasingly targets adaptation, transfer, and continual updat-

ing. A protocol that mixes later recordings into an earlier training set can change both the reported performance level and the apparent ordering of candidate decoders. Quantifying this difference requires paired evaluation at identical test sessions, explicit averaging weights, and a separation between the observed trajectory and the information available before each target.

Building on rolling-origin evaluation and pipeline-level assessment, this study quantifies the difference between retrospective and chronological training information sets at identical BCI test sessions. First, it measures the paired protocol difference across six datasets, 138 participants, and two decoder families. Second, it characterizes variation across participants and recording designs, observed decoder rankings, and a donor-session-count-matched reference. Third, it provides a reporting specification linking each evaluation to its prediction event, available information, temporal unit, and averaging population.

II. RELATED WORK

Open BCI benchmarks standardize datasets, pipelines, and evaluation interfaces, making cross-study comparison increasingly reproducible [1], [2]. Protocol validity remains a separate requirement. Borra *et al.* combine pipeline-wide hyperparameter optimization with repeated random initialization in a within-participant LOSO protocol [3]. Brookshire *et al.* examine leakage in translational EEG studies [4]. The present audit addresses the time at which donor recordings become available while fixing decoder configurations across protocols.

Cross-session motor-imagery methods address nonstationarity through session selection, domain adaptation, and continual updating. Recent examples include relevant-session transfer [5], synchronized self-training [6], nonstationary attention with critic-free adaptation [7], and longitudinal fine-tuning [8]. These methods differ in architecture and adaptation mechanism, but all require an explicit definition of which sessions and labels exist when a prediction is evaluated.

Temporal prediction research supplies that definition. Rolling-origin evaluation places the model at the historical point from which a prediction would have been made [9]; leave-future-out validation and assessment under temporal distribution shift apply the same principle [10], [11]. The present study transfers this principle within longitudinal BCI evaluation and measures the consequence of replacing a completed-trajectory training set with the history actually available at each target.

III. EVALUATION TARGETS AND ESTIMANDS

A. ONE TARGET, TWO INFORMATION SETS

For participant i with ordered sessions $0, \dots, T_i - 1$, the donor sets at target session t are

$$S_{\text{LOSO}}(t) = \{0, \dots, T_i - 1\} \setminus \{t\}, \quad (1)$$

$$S_{\text{EXP}}(t) = \{0, \dots, t - 1\}. \quad (2)$$

Both protocols reserve every trial from session t for testing. Performance is measured by the area under the receiver operating characteristic curve (AUC). For decoder m , the paired contrast is

$$\Delta_{i,t,m} = \text{AUC}_{i,t,m}^{\text{LOSO}} - \text{AUC}_{i,t,m}^{\text{EXP}}. \quad (3)$$

Evaluation begins at $t = 1$, when at least one prior session exists. At the final target, the two donor sets coincide, so $\Delta_{i,T_i-1,m} = 0$. Figure 1 shows the information-set comparison and the two analysis populations.

B. PRIMARY AND SUPPORTING ESTIMANDS

Let $N = 138$ and let m index the two fixed decoder configurations, CSP+LDA and TS+LR. The primary estimand covers every prospective origin in the observed trajectories:

$$\theta_{\text{all}} = \frac{1}{N} \sum_i \frac{1}{2(T_i - 1)} \sum_m \sum_{t=1}^{T_i-1} \Delta_{i,t,m}. \quad (4)$$

The supporting conditional estimand describes targets at which LOSO has at least one later donor:

$$\theta_{\text{future}>0} = \frac{1}{N} \sum_i \frac{1}{2(T_i - 2)} \sum_m \sum_{t=1}^{T_i-2} \Delta_{i,t,m}. \quad (5)$$

Both estimands first average targets and the two decoders within participant and then give participants equal weight. Target-weighted summaries instead give each participant-target pair equal weight after averaging decoders. Dataset-weighted summaries average the six dataset means equally. Writing $\bar{\Delta}_i$ for the within-participant average over targets and both decoders, the final structural zero gives the identity

$$\bar{\Delta}_{i,\text{all}} = \frac{T_i - 2}{T_i - 1} \bar{\Delta}_{i,\text{future}>0}. \quad (6)$$

Thus, all-origin means summarize the complete observed sequence of update decisions, while conditional means summarize decisions with at least one later donor. The scaling factor varies with trajectory length.

IV. METHODS

A. DATASETS AND TIME UNITS

The analysis uses the fixed collection of six datasets in Table 1. Each provides binary motor-imagery labels, at least three source-defined ordered units per participant, chronological unit identifiers, and complete evaluation coverage for both classical decoders. The inclusion manifest records the participants and targets represented in the result files. This curated collection defines the empirical scope of the audit, comprising 138 participants.

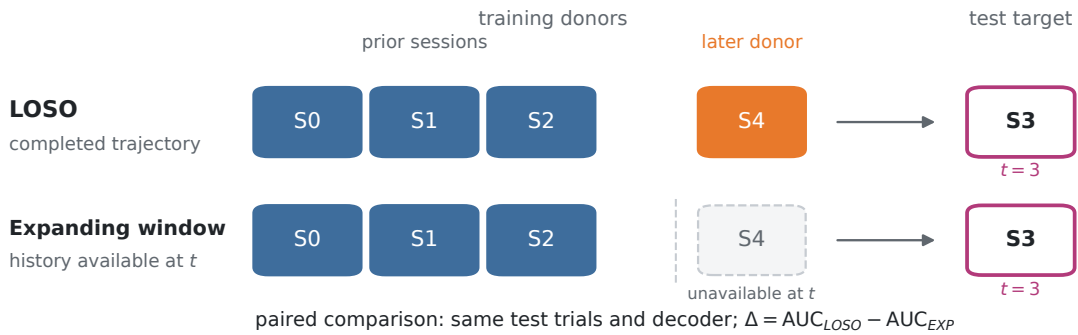
Zhou2016 and Zhou2020 are distinct datasets [12], [13]. Ma2020 records right-hand and right-elbow imagery [14]; the remaining datasets use left- and right-hand imagery. The source-defined unit is a session or recording day except in the primary Ma2020 analysis, which uses the 15 released block identifiers. A sensitivity analysis regroups those blocks into their three recording days. Session-index lag is interpreted within dataset because a one-unit step does not represent a common elapsed time across studies.

Every recording was band-pass filtered from 8 to 35 Hz, resampled to 128 Hz, and epoched over $[0, 3)$ s relative to the task cue, giving 384 samples per trial. Ma2020 CNT files were loaded after removing horizontal and vertical electroculography channels, the mastoid reference, and optional electromyography channels, leaving 62 scalp EEG channels. Preprocessing did not use participant-wide or cross-session normalization, and its parameters did not depend on target-session labels or performance.

B. DECODERS AND INFORMATION ISOLATION

The paired audit used two established decoder families with different feature geometries and complete coverage across all six datasets. Common spatial patterns with linear discriminant analysis (CSP+LDA) represent a supervised spatial-filter pipeline; tangent-space features with logistic regression (TS+LR) represent covariance geometry in a tangent space.

(a) Donor availability at the same target



(b) Analysis populations

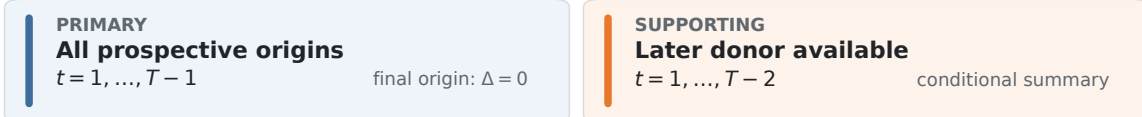


FIGURE 1. Information sets for the paired evaluation. (a) At the same early target, LOSO trains on prior sessions and a later session, whereas the expanding window uses only the history already available. The test trials and decoder are identical. (b) The primary estimand averages all prospective origins, including the final origin where the donor sets coincide; the supporting estimand conditions on the availability of a later donor.

TABLE 1. Longitudinal motor-imagery datasets in the audit. Channel counts refer to EEG channels retained by the audited loader.

Dataset	Participants	Ordered units	EEG channels	Source
BNCI2014_004	9	5 sessions	3 bipolar	[15]
Zhou2016	4	3 sessions	14	[12]
Zhou2020	20	6 or 7 sessions	41 or 26	[13]
Kumar2024	18	6 sessions	22	[16]
Ma2020	25	15 blocks in 3 days	62	[14]
Stieger2021	62	7 or 11 sessions	60 retained	[17]

Their deterministic, fixed configurations keep the comparison centered on evaluation protocol rather than decoder search.

CSP+LDA used $\min(8, p)$ common spatial pattern (CSP) components for p channels, reduced to the next lower even number when required. CSP used Ledoit–Wolf regularization, and linear discriminant analysis used the `lsqr` solver with automatic shrinkage [18]. TS+LR estimated a Ledoit–Wolf covariance for each trial, learned the affine-invariant tangent-space reference from the donor covariances, and fitted L_2 logistic regression with $C = 1$ and the `lbfgs` solver [19]. All hyperparameters were held fixed across protocols.

Each participant, target, protocol, and decoder received a fresh fit. CSP filters, LDA parameters, the tangent-space reference, and logistic-regression parameters were estimated only from donor trials. The target session contributed final scores only. The inclusion audit verified complete participant–target fits, unique result keys, and AUC values in the valid range. AUC was the primary endpoint; the Ma2020 day-level sensitivity also reported balanced accuracy.

C. STATISTICAL ANALYSIS

The primary analysis used participant-level all-origin means, two-sided one-sample t intervals, and Wilcoxon signed-rank sensitivity tests. The same procedures were applied to the conditional estimand. Target-weighted intervals used 5,000 participant-cluster bootstrap resamples. Dataset means were summarized with participant-level intervals and equal-dataset weighting.

A random-effects synthesis used each dataset’s participant-level mean and sampling variance. Between-dataset variance was estimated by restricted maximum likelihood. The summary interval used a t reference across the six datasets, and the prediction interval incorporated mean-estimation uncertainty and between-dataset variance. Heterogeneity was summarized by I^2 . Exact interval formulas and a residual-scale sensitivity analysis are given in Appendix C. The intervals describe estimation uncertainty without multiplicity adjustment; secondary contrasts and sensitivity checks are interpreted descriptively.

Protocol sensitivity was evaluated from the participant distribution of Δ and the higher-scoring decoder by participant

mean AUC under each protocol. We recorded the direction of each observed ranking change and its persistence when both within-protocol decoder differences exceeded 0.005, 0.01, or 0.02 AUC. These thresholds describe proximity to a tie. Two additional AUC thresholds, 0.70 and 0.75, were retained as illustrative sensitivity checks in the appendix; they were not treated as clinical cutoffs.

D. COMMON-SUPPORT AND ROBUSTNESS ANALYSES

A uniform donor-session-count-matched reference sampled t sessions from the LOSO-eligible donor pool and averaged five draws from a fixed random seed. The main decomposition uses all conditional targets. A second decomposition uses strict-future common support, $T_i - 1 - t \geq t$, to align with an additional arm trained exclusively on later sessions. On each support, the contrast is partitioned as

$$\begin{aligned} \text{AUC}_{\text{LOSO}} - \text{AUC}_{\text{EXP}} = & (\text{AUC}_{\text{LOSO}} - \text{AUC}_{\text{matched}}) \\ & + (\text{AUC}_{\text{matched}} - \text{AUC}_{\text{EXP}}). \end{aligned} \quad (7)$$

The first term compares the full donor pool with the matched reference; the second compares that reference with the t prior sessions. Matching session count leaves trial count and session characteristics dependent on the sampled donors. The components form an additive description under this specific reference and support. Participant intervals condition on the realized draws. Monte Carlo diagnostics use support-specific weights and a covariance-free bound based on estimated marginal variances (Appendix C).

Two additional operational policies trained on the immediately previous session or the first session only. The target-position analysis related the conditional contrast to the number of later sessions available to LOSO within each dataset. An adjacent-session drift proxy used the affine-invariant Riemannian distance between unconditional mean trial covariance matrices at $t - 1$ and t , standardized within dataset. Its target-level regression adjusted for target index, decoder, and dataset, with participant-clustered standard errors. Finally, the Ma2020 day-level analysis concatenated blocks 1–5, 6–10, and 11–15, retained all trials, and refitted both decoders.

V. RESULTS

A. LOSO REPORTS A HIGHER PERFORMANCE LEVEL

Across all prospective origins, mean AUC was 0.768 (95% CI [0.747, 0.789]) under LOSO and 0.738 (95% CI [0.718, 0.759]) under expanding-window validation. Their paired participant-weighted difference was +0.030 AUC (95% CI [+0.025, +0.034]). Target weighting produced a similar estimate. When the final convergence point was removed, the conditional difference was +0.035 AUC (95% CI [+0.030, +0.040]). Figure 2 shows the corresponding participant distributions, and Table 2 reports the aggregation analyses.

The two decoder families showed the same protocol ordering. The all-origin mean difference was +0.031 AUC for CSP+LDA and +0.029 for TS+LR. Under the primary estimand, LOSO produced the larger participant mean for 89.9%

of participants, and the difference exceeded 0.02 AUC for 66.7%. The higher-scoring decoder differed between protocols for 14 of 138 participants (10.1%). Of these, 10 changed from CSP+LDA under LOSO to TS+LR under expanding-window evaluation, and 4 changed in the opposite direction. Only 1 participant had a ranking change with decoder differences exceeding 0.01 AUC under both protocols. The corresponding conditional count was 6 of 19 changes. TS+LR had the higher dataset-average AUC in all six datasets under both protocols and both estimands. Figure 3 distinguishes observed ranking changes from near ties.

B. THE MAGNITUDE VARIES ACROSS RECORDING DESIGNS

Equal weighting of the six datasets gave an all-origin mean of +0.024 AUC. The random-effects estimate was +0.024 (95% CI [+0.009, +0.039]), with $I^2 = 83.7\%$ and a prediction interval of [-0.016, +0.064]. Thus, the average protocol separation was positive across the observed designs. The prediction interval crosses zero, representing uncertainty in both direction and magnitude for another recording design.

Figure 4 displays the all-origin estimates by dataset and decoder. The dataset means were positive in all six datasets. Uncertainty was widest for the cohorts with four and nine participants.

C. DECOMPOSITION UNDER A UNIFORM DONOR-SESSION REFERENCE

On all 1,001 conditional participant–target pairs, the total protocol difference was +0.035 AUC. The full-pool minus matched-reference component was +0.025, and the matched-reference minus prior-history component was +0.009. Both comparisons use the same targets and participant weights as the conditional estimand. The reference matches the number of donor sessions; variation in their trial counts remains within the sampled composition.

Restricting all arms to the 560 early targets with sufficient strictly future donors gave a total difference of +0.053 AUC, partitioned into +0.040 and +0.013. Training exclusively on the matched number of future sessions instead of prior sessions gave a difference of +0.017. Table 3 reports each support separately. The covariance-free plug-in standard-error bounds for the sampled reference were 0.0180 and 0.0221 AUC on these two supports. They use estimated marginal variances; the joint Monte Carlo standard error is unidentified from the retained summaries. Appendix C specifies the diagnostic separately from participant uncertainty.

D. TEMPORAL-UNIT AND DRIFT ANALYSES

For Ma2020, the all-origin difference was +0.033 AUC when the released blocks were treated as ordered units and +0.019 after the same trials were grouped into three recording days. The conditional differences were +0.036 and +0.039 AUC, respectively; the day-level difference was also positive for balanced accuracy. Equation (6) gives an all-origin scaling

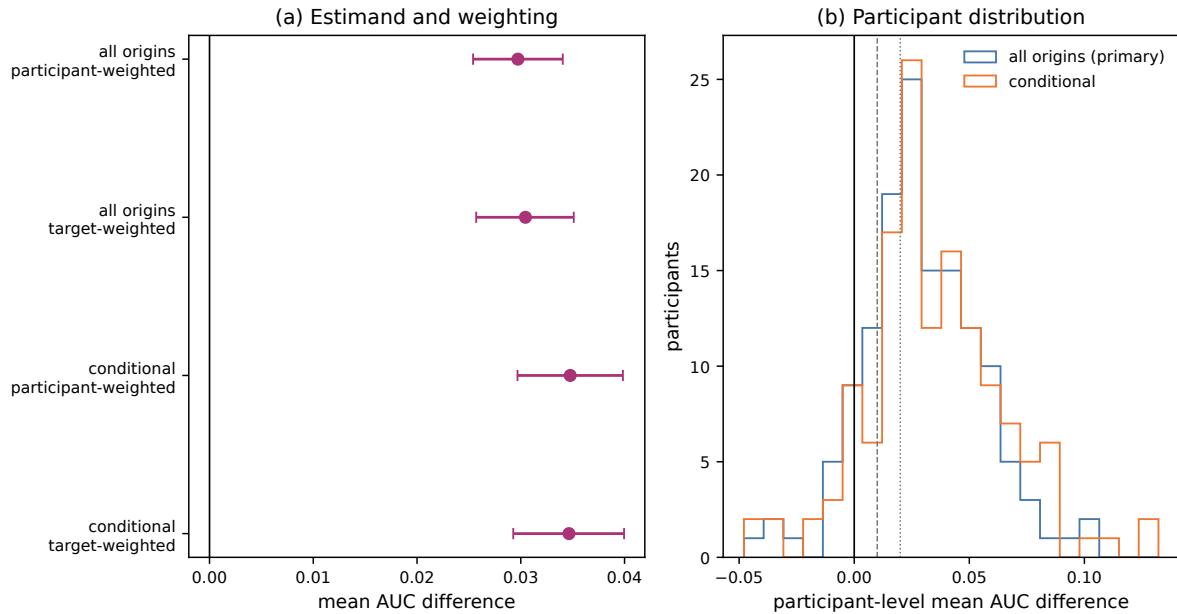


FIGURE 2. Primary and supporting estimates. (a) Participant- and target-weighted LOSO minus expanding-window AUC. (b) Participant-level distributions. The vertical reference lines mark zero, 0.01, and 0.02 AUC.

TABLE 2. LOSO minus expanding-window AUC under the primary all-origin estimand and the conditional summary. The all-origin estimand includes the final structural zero. Conditional estimates require at least one later session. Target-weighted intervals use a participant-cluster bootstrap.

Estimand and aggregation	Units	Mean difference [95% CI]
All origins, participant-weighted	138	+0.030 [+0.025, +0.034]
All origins, target-weighted	138	+0.030 [+0.026, +0.035]
All origins, dataset-weighted	6	+0.024 [+0.008, +0.040]
All origins, random-effects	6	+0.024 [+0.009, +0.039]
Conditional, participant-weighted	138	+0.035 [+0.030, +0.040]
Conditional, target-weighted	138	+0.035 [+0.029, +0.040]
Conditional, dataset-weighted	6	+0.030 [+0.012, +0.047]
Conditional, random-effects	6	+0.030 [+0.014, +0.046]

TABLE 3. Reference-dependent descriptive decomposition. Uniform donor-session-count matching is analyzed on the full conditional support and, separately, on the strict-future support required by the future-only reference. The two uniform-reference components sum to the total contrast within each support. Matching session count does not match trial count.

Contrast	Participants	Mean AUC difference [95% CI]
<i>Later donor available: 1,001 participant-targets</i>		
LOSO — expanding window	138	+0.0348 [+0.0297, +0.0398]
LOSO — donor-count-matched reference	138	+0.0255 [+0.0219, +0.0291]
Donor-count-matched reference — expanding window	138	+0.0093 [+0.0049, +0.0137]
<i>Strict-future common support: 560 participant-targets</i>		
LOSO — expanding window	138	+0.0529 [+0.0452, +0.0607]
LOSO — donor-count-matched reference	138	+0.0401 [+0.0349, +0.0453]
Donor-count-matched reference — expanding window	138	+0.0128 [+0.0058, +0.0198]
Strict future — expanding window	138	+0.0165 [+0.0073, +0.0258]

factor of 13/14 for blocks and 1/2 for days. Thus, the all-origin comparison combines a change in the temporal unit with a change in the weight of the final structural zero. The conditional comparison retains a positive separation under both time units. Table 7 also reports the immediately-previous-session and first-session policy contrasts.

The target-position curves differed by dataset and were largest at early positions in several longer trajectories. The adjusted coefficient for the adjacent unconditional covariance-distance proxy was near zero. These analyses are shown with their support and influence checks in Appendix A.

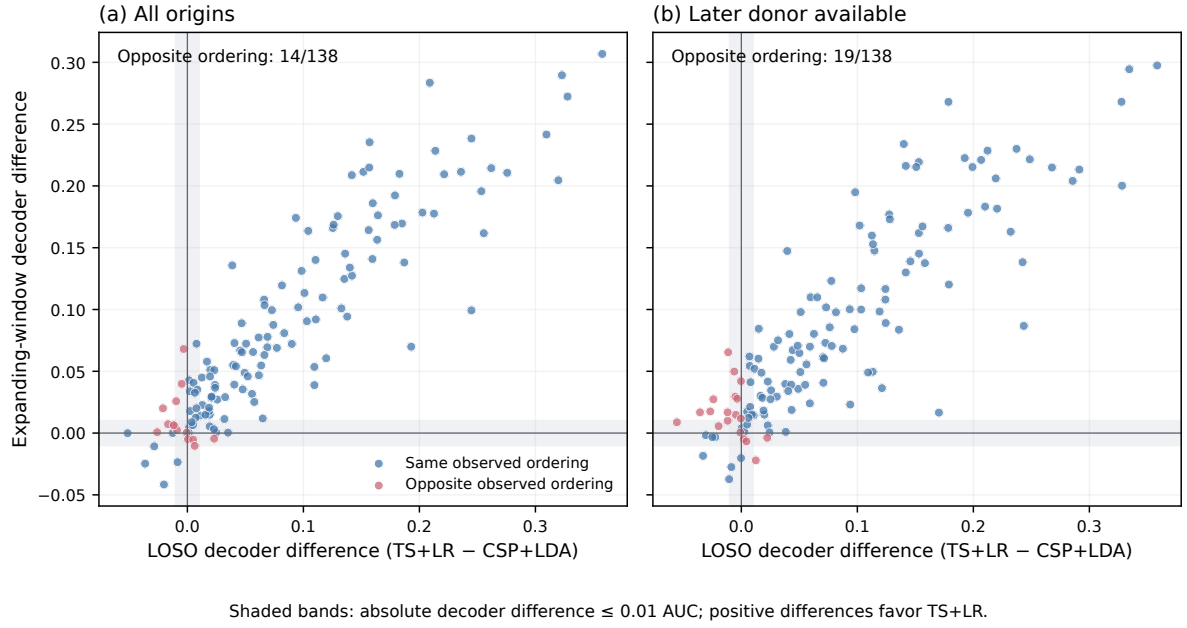


FIGURE 3. Participant-level decoder margins under the two protocols. Each point is one participant; positive margins favor TS+LR over CSP+LDA. Opposite-sign quadrants indicate a change in the observed higher-scoring decoder. Shaded bands mark absolute margins at or below 0.01 AUC; the threshold is a descriptive sensitivity check.

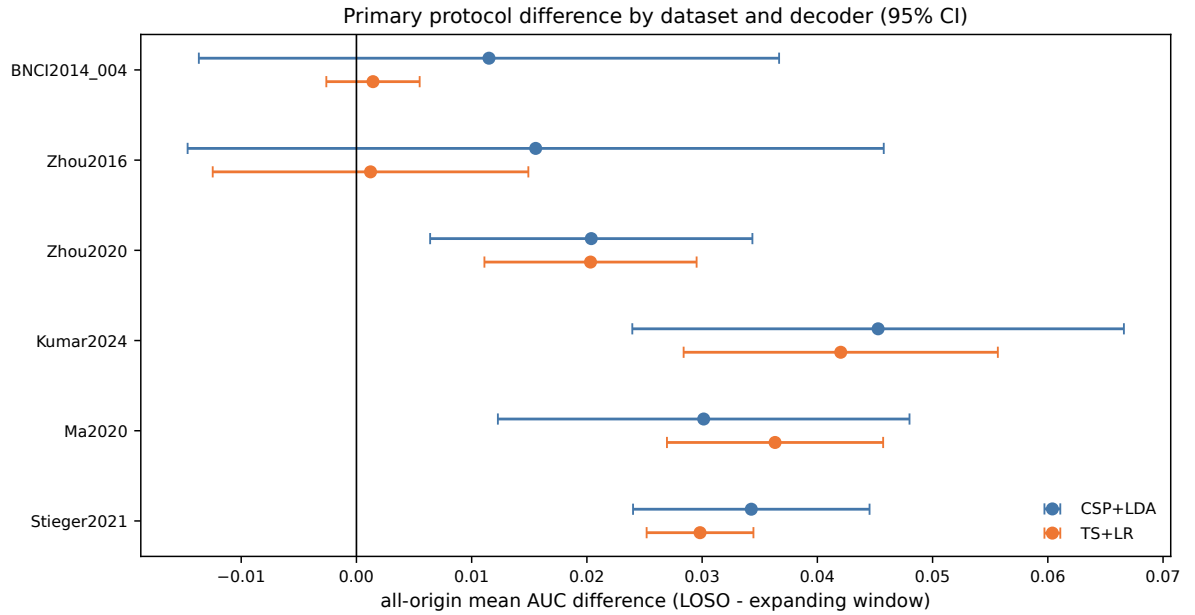


FIGURE 4. All-origin LOSO minus expanding-window AUC by dataset and decoder. Points are dataset means of participant-level contrasts, and bars are 95% confidence intervals across participants.

VI. DISCUSSION

A. THE SPLIT DEFINES THE EVALUATION TARGET

The main result quantifies how the training information set changes the measured performance at an identical target. LOSO measures decoding after the completed participant trajectory is available. Expanding-window validation measures decoding from the labeled history available before each

target. The same participant, target trials, preprocessing, and decoder were paired throughout; the designed difference was the training information set.

The primary all-origin estimate answers the longitudinal decision-sequence question over the recorded observation period. The conditional estimate summarizes targets where LOSO has later donors. Equation (6) connects these quan-

ties and shows how the observed endpoint and trajectory length affect all-origin averages. Reporting both preserves the distinction between the full decision sequence and the subset with different training information.

B. CONSEQUENCES FOR BENCHMARKING AND ADAPTATION

The protocol changed the pooled performance level and the observed higher-scoring decoder for a subset of participants. Most all-origin ranking changes involved decoder scores within 0.01 AUC under at least one protocol. These results describe the sensitivity of observed rankings to the split; a prospective selection policy would additionally define how a decoder is chosen from available history and evaluate that choice on subsequent sessions. Reporting the decoder margin alongside the winner count separates near ties from larger ranking changes.

Under the uniform donor-session reference, the full-pool versus matched-reference component was larger than the matched-reference versus prior-history component. The latter holds session count fixed while changing donor identity, trial count, and session characteristics. This reference-dependent description identifies which comparisons differ in score. Session-selection and adaptation studies can use the same construction to state precisely which donor pool each reported score represents.

Participant- and target-weighted summaries were closely aligned, whereas the random-effects analysis showed substantial cross-dataset heterogeneity. The result shows a positive mean in each of the six observed datasets and a design-specific magnitude. The random-effects prediction interval includes both positive and negative values. Dataset-level effects and the prediction interval therefore remain part of the interpretation of the pooled estimate.

C. REPORTING PROCEDURE AND EMPIRICAL SCOPE

A longitudinal evaluation can be specified by the five reporting fields in Table 4. A completed-trajectory question maps to LOSO; chronological offline retraining maps to an expanding window. When both questions matter, both estimates can be reported and named by their information sets.

The empirical estimates describe binary motor imagery, two fixed classical decoder configurations, and the six recording designs in Table 1. The evaluated update policy refits from fully labeled prior sessions and measures discrimination by AUC on fixed recorded trajectories. Neural decoders, multiclass outcomes, delayed labels, and closed-loop adaptation define additional empirical settings. Feedback and user learning, as studied in closed-loop BCI [20], add interactions between the decoder and the recorded trajectory. Their evaluation requires specifying the corresponding update events and information availability. Alternative donor-sampling references and outcome metrics likewise define additional comparisons.

VII. CONCLUSION

LOSO and expanding-window validation estimate different longitudinal BCI targets. Across six motor-imagery datasets, LOSO reported an average all-origin AUC 0.030 higher, with variation across recording designs. Observed participant-level decoder rankings changed mainly where the scores were close. A uniform donor-session-count reference provided an additive description of the conditional difference on explicit support. Longitudinal benchmarks should define the evaluation decision, match its training information, and report its temporal unit and averaging weights.

DATA AVAILABILITY

This study is a secondary analysis of the datasets cited in Table 1. Access conditions and raw-data licenses are governed by the original repositories and dataset publications. The analysis uses no newly collected participant data.

CODE AVAILABILITY

The accompanying review supplement provides the manuscript snapshot, analysis code, version-pinned result-analysis dependencies, dataset registry, inclusion and support manifests, result-level CSV files, seeds, historical Stieger reuse provenance, SVG figure sources, matched vector-PDF figures, and LaTeX source. Its result-only workflow regenerates the analyses and figures without raw EEG. The submitted-result audit checks result coverage, duplicate keys, score ranges, and final-origin structural zeros. Raw-cache checks, including epoch length and retained channels, are a separate workflow requiring the original recordings or local caches.

ETHICS STATEMENT

The analysis uses previously released, de-identified datasets and collects no new human-participant data. Ethical approvals and consent procedures are reported in the original dataset publications.

AUTHOR CONTRIBUTIONS

Yiming Shen: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing—original draft, and writing—review and editing.

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APPENDIX A ROBUSTNESS AND INTERPRETATION CHECKS

A. PARTICIPANT-LEVEL RANKING AND THRESHOLD SENSITIVITY

The primary participant-level median was +0.026 AUC (IQR [+0.014, +0.046]). Table 5 reports the prevalence of positive

TABLE 4. Reporting specification for longitudinal decoding evaluation, with this audit as a worked example.

Reporting field	Specification in this audit
Prediction and update event	A fresh decoder fit before each target session.
Available recordings and labels	Expanding window: all fully labeled prior sessions. LOSO: all fully labeled sessions other than the target.
Target contributions to fitting	No target contributions to learned parameters; fixed preprocessing followed by scoring on the target trials.
Temporal unit and endpoint	Source-defined session or block, through the final recorded unit; Ma2020 additionally regrouped by recording day.
Averaging population	Targets and two decoders averaged within participant, then participants weighted equally. The all-origin average includes the final structural zero; conditional, target-weighted, and dataset-weighted summaries are identified separately.

TABLE 5. Observed participant-level magnitude and counts, n (%), of the protocol difference.

Metric	All origins	Conditional
Median [IQR]	+0.026 [+0.014, +0.046]	+0.031 [+0.017, +0.053]
Participants with $\Delta > 0$	124 (89.9%)	124 (89.9%)
Participants with $\Delta > 0.01$	113 (81.9%)	115 (83.3%)
Participants with $\Delta > 0.02$	92 (66.7%)	98 (71.0%)

TABLE 6. Observed participant-specific decoder ordering and illustrative threshold crossings ($n = 138$). Arrows go from LOSO to expanding-window ordering. Bilateral margins require both absolute TS+LR-minus-CSP+LDA mean AUC differences to exceed the threshold. Threshold crossings average both decoders and targets within participant. These are descriptive point-estimate comparisons.

Outcome	All origins	Conditional
Opposite decoder ordering	14 (10.1%)	19 (13.8%)
CSP+LDA $\rightarrow$ TS+LR	10	15
TS+LR $\rightarrow$ CSP+LDA	4	4
Flip, both margins > 0.005	7	11
Flip, both margins > 0.01	1	6
Flip, both margins > 0.02	1	1
LOSO-only AUC ≥ 0.70	13	16
Expanding-only AUC ≥ 0.70	1	1
LOSO-only AUC ≥ 0.75	22	31
Expanding-only AUC ≥ 0.75	0	0

differences and differences exceeding the stated thresholds. Table 6 reports observed ranking changes, margin sensitivity, and threshold crossings in both directions.

B. TARGET POSITION AND LATER-SESSION AVAILABILITY

Figure 5 displays the conditional difference by the number of later sessions available only to LOSO. Because this count equals $T_i - 1 - t$, each curve jointly reflects target position, accumulated history, and trajectory length.

C. MA2020 TEMPORAL UNIT

Treating Ma2020's 15 blocks as ordered units gave conditional and all-origin differences of +0.036 and +0.033 AUC. Grouping the same trials into three recording days gave +0.039 and +0.019, respectively (Figure 6). On the day-level conditional support, the balanced-accuracy difference was

TABLE 7. Alternative training policies and Ma2020 time-unit sensitivity.

Analysis	n	Mean AUC difference [95% CI]
LOSO — immediately previous	138	+0.066 [+0.058, +0.074]
LOSO — first session	138	+0.103 [+0.091, +0.116]
Ma2020 block, conditional	25	+0.036 [+0.024, +0.048]
Ma2020 day, conditional	25	+0.039 [+0.020, +0.058]
Ma2020 block, all origin	25	+0.033 [+0.022, +0.044]
Ma2020 day, all origin	25	+0.019 [+0.010, +0.029]

+0.033 (95% CI [+0.019, +0.047]).

D. ADJACENT COVARIANCE-DISTANCE PROXY

The participant-clustered coefficient for standardized adjacent unconditional covariance distance was -0.0001 AUC (95% CI [-0.0050, +0.0047], $p = 0.9530$). Ninety-one of 2,002 target-decoder observations exceeded the Cook's-distance threshold; removing them produced a similarly near-zero association.

APPENDIX B OBSERVED SUPPORT AND ALTERNATIVE POLICIES

The strict-future arm requires $t \leq (T_i - 1)/2$ and therefore represents early targets. Table 8 reports its participant-target support and dataset composition. The uniform reference is defined on all conditional targets; the future-only reference additionally requires this early-target support.

The released chronology consists of five sessions for BNCI2014_004, three for Zhou2016, six or seven for Zhou2020, six recording days for Kumar2024, 15 blocks over three days for Ma2020, and seven or 11 recording days for Stieger2021. Exact elapsed intervals were unavailable for BNCI2014_004; all temporal contrasts therefore use source-defined order rather than a pooled elapsed-time scale.

APPENDIX C INTERVAL DEFINITIONS AND DONOR-SAMPLING UNCERTAINTY

A. DATASET-LEVEL SYNTHESIS

Let y_d and v_d be the participant mean and its estimated sampling variance for dataset d , and let $k = 6$. With restricted-maximum-likelihood estimate $\hat{\tau}^2$, the weights and pooled mean are

$$w_d = (v_d + \hat{\tau}^2)^{-1}, \quad \hat{\mu} = \frac{\sum_d w_d y_d}{\sum_d w_d}. \quad (8)$$

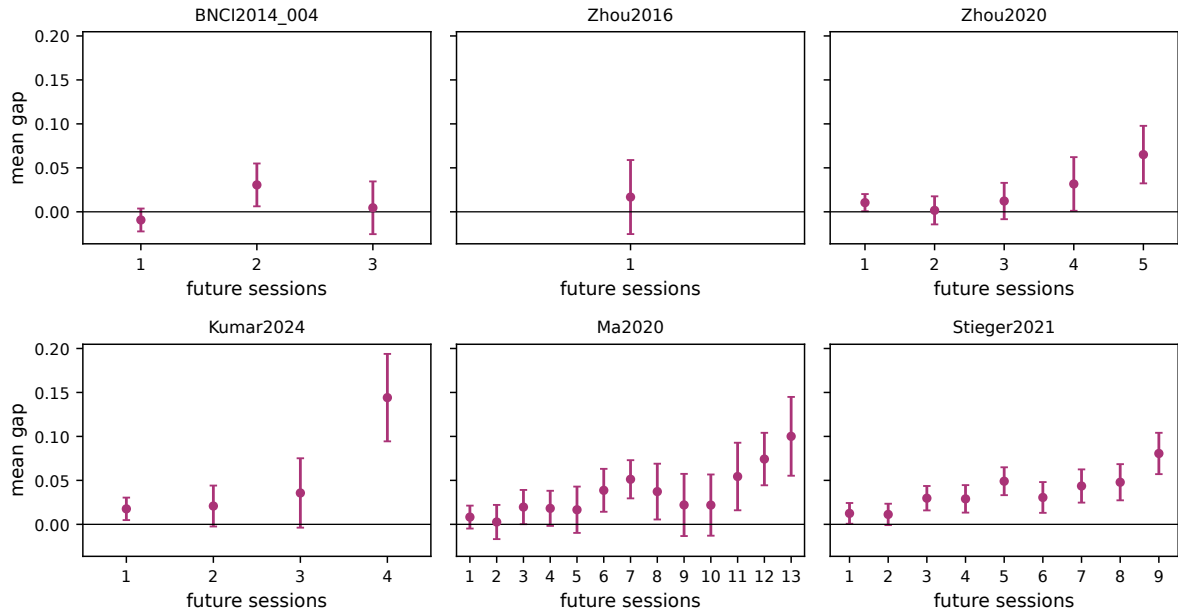


FIGURE 5. Dataset-specific association between the protocol difference and the number of later sessions available to LOSO. Each point averages participants and both decoders at a common within-dataset index; bars are 95% participant-level confidence intervals.

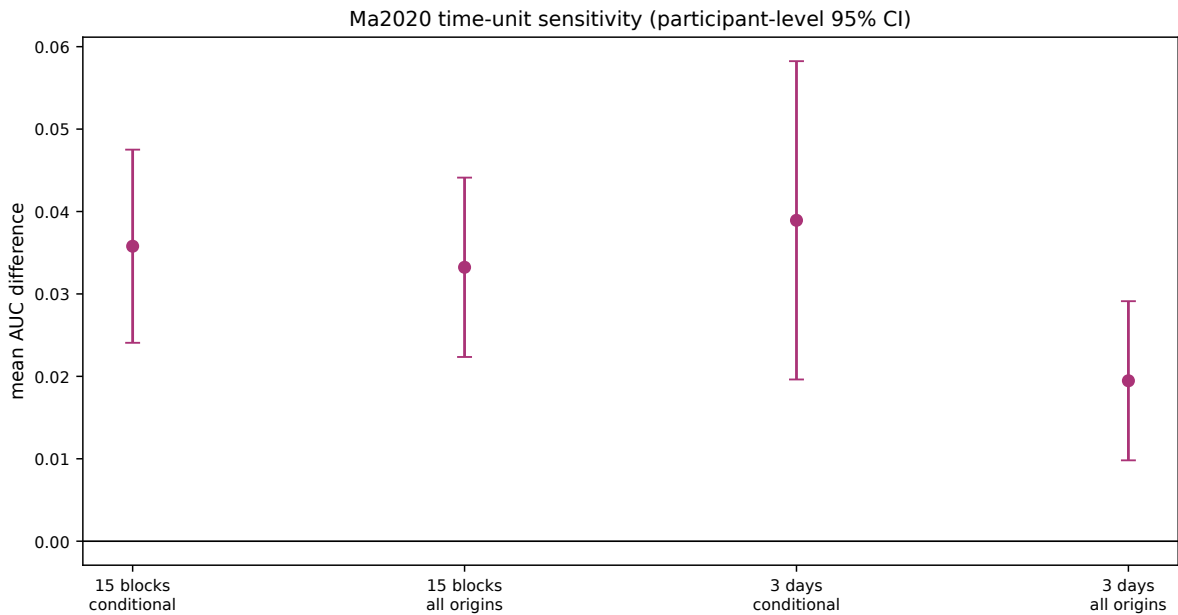


FIGURE 6. Ma2020 sensitivity to the temporal unit. The block analysis uses 15 ordered blocks; the day analysis concatenates five consecutive blocks within each of three recording days. Bars are 95% participant-level confidence intervals.

The reported interval is $\hat{\mu} \pm t_{k-1,0.975}(\sum_d w_d)^{-1/2}$; the prediction interval is $\hat{\mu} \pm t_{k-2,0.975}\sqrt{\hat{\tau}^2 + (\sum_d w_d)^{-1}}$. As a residual-scale sensitivity analysis, define $q = \sum_d w_d(y_d - \hat{\mu})^2/(k-1)$ and replace the mean standard error by $\sqrt{q/\sum_d w_d}$. This gives intervals of $[+0.009, +0.039]$ for all origins and $[+0.014, +0.047]$ for conditional targets.

B. FIVE-DRAW UNIFORM REFERENCE

The cached per-row standard deviation $s_{r,0}$ uses divisor $B = 5$. The sample-standard-deviation correction gives a marginal standard error of $e_r = s_{r,0}/\sqrt{B-1}$ for the sampled row mean. Rows share random draws across decoders; the cached summaries retain marginal variation but not the cross-row covariances. For nonnegative participant-averaging weights

TABLE 8. Observed participant–target support by dataset. The two decoders share these targets. The uniform donor-count-matched reference uses the conditional support; the future-only reference requires strict-future support, $t \leq (T - 1)/2$. Mean donor-session count is the target-weighted expanding-window count $\bar{t}$.

Dataset	Participants	All origins	Conditional	Strict future
BNCI2014_004	9	36	27	18
Zhou2016	4	8	4	4
Zhou2020	20	119	99	59
Kumar2024	18	90	72	36
Ma2020	25	350	325	175
Stieger2021	62	536	474	268
Total	138	1139	1001	560
Mean donor-session count	—	5.36	4.96	2.93

a_r summing to one, we report

$$U = \sum_r a_r e_r. \quad (9)$$

The triangle inequality bounds the aggregate standard error by U under the estimated marginal variances, allowing arbitrary cross-row covariance. This is a plug-in bound rather than a statistical confidence upper limit. The corresponding values are 0.0180 AUC on all conditional targets and 0.0221 on strict-future support. The result files also report the median, 95th percentile, and maximum marginal row-level standard error. Participant confidence intervals for the partition condition on the realized donor draws; donor-sampling uncertainty is reported separately. The unsampled LOSO–expanding contrast is unchanged by the reference draws.

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